QARGO - QR Assisted Robotic Goods Operator

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INTRODUCTION

Autonomous material-handling robots improve efficiency, accuracy, and safety in warehouses and production systems. In many industries, manual handling and transportation of heavy goods are physically demanding, time-consuming, and prone to human error, creating the need for safer and more automated material-handling solutions. Vision-guided forklift research has demonstrated automatic pallet engagement using camera-based guidance [1], while QR-based goods-segregation robots have combined web-camera scanning, line following, IR sensing, and forklift mechanisms to identify goods and transport them to designated locations [2]. Inspired by these approaches, our project QARGO (QR Assisted Robotic Goods Operator) uses a 6-channel IR sensor array for guided navigation, a Raspberry Pi 4B and camera for QR-code identification and destination selection, and an Arduino Mega 2560 for motor, sensor, and fork-lifter control. The system autonomously picks, identifies, transports, and delivers blocks to their assigned docking stations.

Key Working Principle:

OBJECTIVES

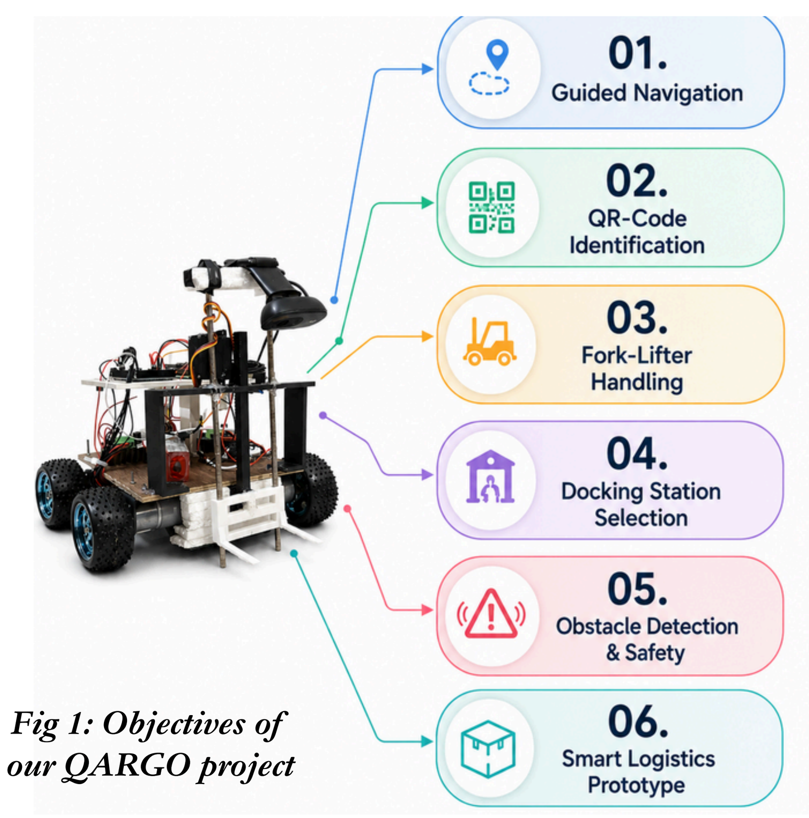

Fig 1: Objectives of our QARGO project

SCHEMATIC DIAGRAMS

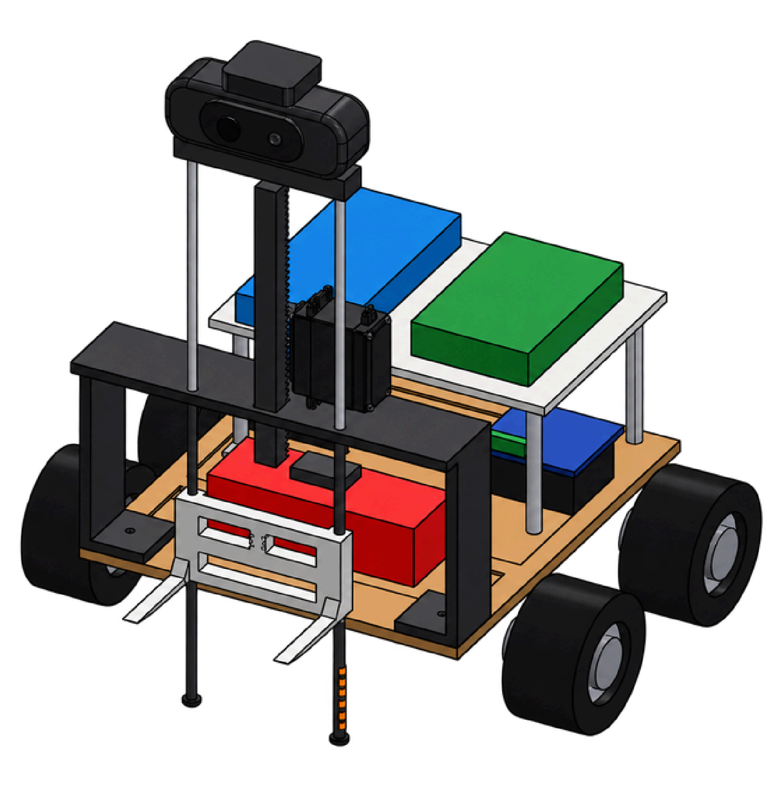

Fig 2: CAD Model of QR Assisted Robotic Goods Operator

METHODOLOGY

The QARGO system was developed through a stepwise mechatronic process integrating guided navigation, QR-based decision making, and automated material handling.

- System & Chassis Design
- Component Integration
- IR Line Following
- QR Code Scanning
- Dock Selection
- Fork-Lifter Control
- Testing & Validation

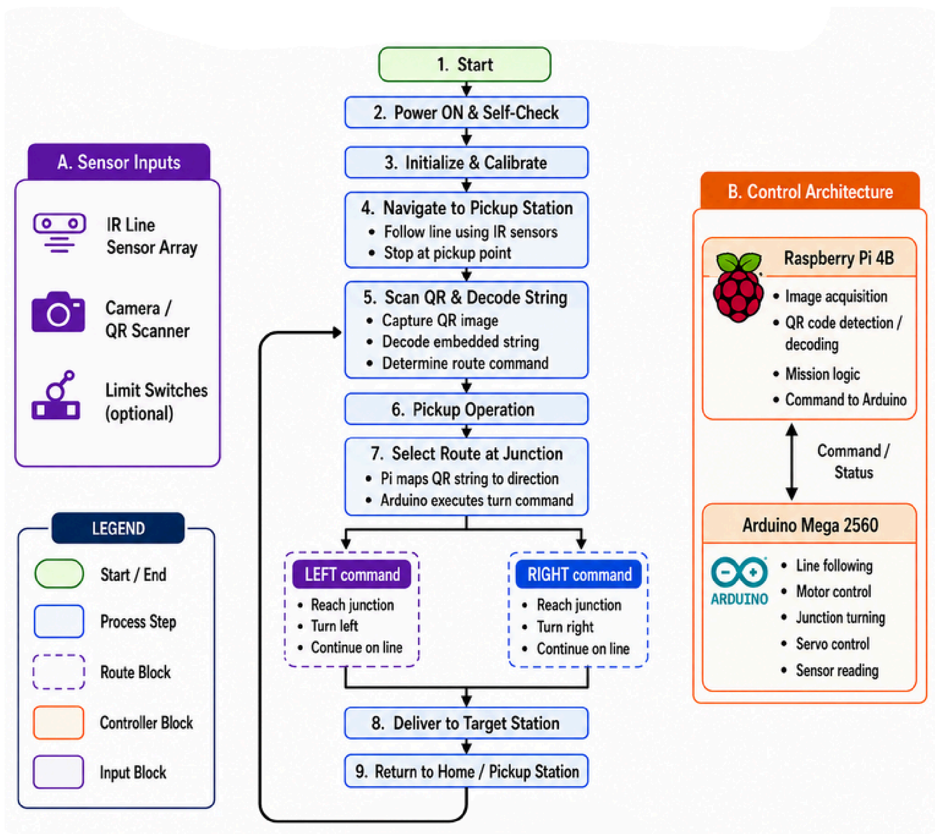

Fig 3: System Flowchart

- **Chassis:** Modular frame supporting electronics, drive system, camera, sensors, and fork-lifter.
- **6-Channel IR Sensor Array:** Detects the black line and provides feedback for guided navigation.
- **Camera / Webcam:** Scans QR codes for block identification and destination selection.
- **Raspberry Pi 4B:** Handles QR decoding, destination mapping, mission logic, and Arduino communication.
- **Arduino Mega 2560:** Controls line following, motors, servos, sensors, and safety functions.
- **Drive Motors:** Four 12 V, 130 RPM gear motors provide stable four-wheel movement.
- **Motor Drivers**: BTS motor driver controls high-current motor operation.
- **Fork-Lifter Mechanism**: MG996R high-torque servo performs lifting and lowering; SG90 assists block release.
- **Power Supply:** 3S 11.1 V LiPo battery supplies power through 2 buck converters: 1× LM2596 and 1× XL4015 for regulated voltage distribution.

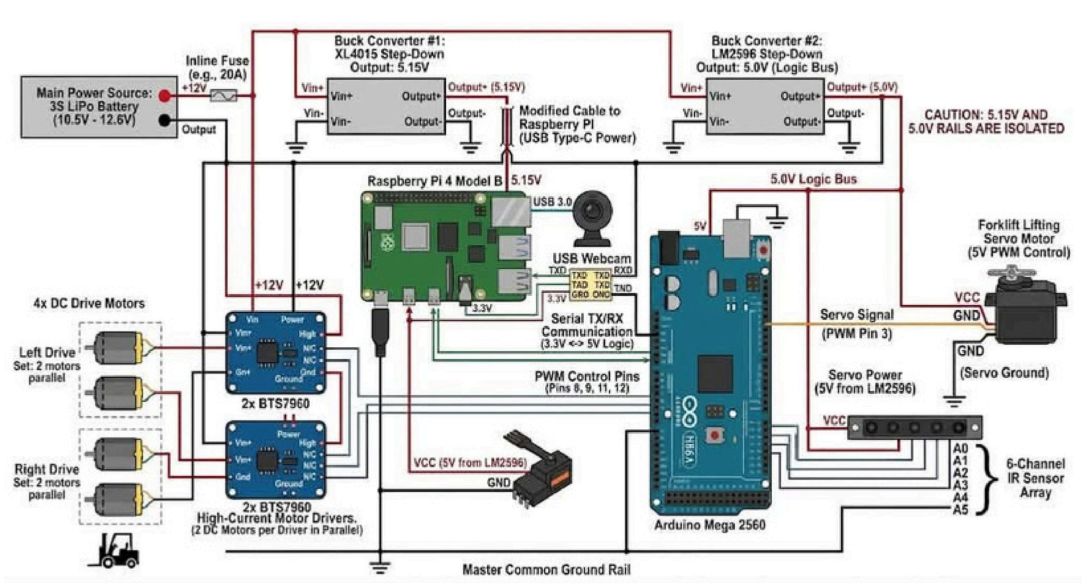

Fig 4: Circuit Diagram

RESULTS & DISCUSSION

The developed QARGO prototype successfully integrates the major subsystems required for autonomous material handling: guided line navigation, QR-code identification, destination selection, obstacle sensing, and fork-lifter actuation. The 6-channel IR sensor array provides line-following feedback, while the Raspberry Pi processes QR information and the Arduino Mega controls the motors and actuators.

During integration, system performance was found to depend strongly on line-sensor calibration, QR visibility, camera alignment, fork positioning, and stable power supply. QR codes must remain clearly visible during scanning, while accurate pickup requires proper alignment between the fork and wooden block. Reliable serial communication between the Raspberry Pi and Arduino is also essential for correct docking decisions.

Conclusions:

- Integrated autonomous material handling achieved.
- QR-based dock selection implemented.
- IR-guided navigation established.
- Fork-lifter mechanism integrated.
- Calibration and mechanical alignment affect reliability.
- Further repeated trials are required for quantitative performance evaluation.
- Coordination and synchronization among multiple QARGO robots operating simultaneously is yet to be implemented.

FUTURE RESEARCH & DEVELOPMENT

01 Multi-Robot Coordination
Develop synchronization for multiple QARGO robots working together.

02 Dynamic Path Planning
Implement real-time route optimization and traffic avoidance.

03 AI & Machine Learning
Improve QR recognition and decision making with AI.

04 Advanced Sensors
Add depth camera / LiDAR for better obstacle detection.

05 Cloud Integration
Enable data logging, remote monitoring and control.

06 Battery Optimization
Increase battery life and add auto-charging feature.

REFERENCES

[1] M. Seelinger and J.-D. Yoder, "Automatic visual guidance of a forklift engaging a pallet," Robotics and Autonomous Systems, Vol. 54, No. 12, pp. 1026–1038, 2006.

[2] K. Bhavana, B. Lalitha, T. Srinivasa Rao, and M. V. Ramesh, "Goods Segregation Robot for Industry," International Journal of Emerging Trends in Engineering Research, Vol. 8, No. 9, pp. 5828–5831, 2020.